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A COVQ-Based Image Coder for Channels with Bit Errors and Erasures

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Abstract—We illustrate how channel optimized vector quantization (COVQ) can be used for channels with *both* bit-errors and bit-erasures. First, a memoryless channel model is presented, and the performance of COVQ's trained for this channel is evaluated for an i.i.d. Gaussian source. Then, the new method is applied in implementing an error-robust sub-band image coder, and we present image results that illustrate the resulting performance. Our experiments show that the new approach is able to outperform a traditional scheme based on separate source and channel coding.

Index Terms—Binary erasure channels, binary symmetric channels, channel-optimized quantization, image coding, joint source-channel coding.

I. INTRODUCTION

FOR packet data networks where parts of the overall transmission are over wireless links, phenomena occur that are not present in traditional wired networks (over optical fiber and/or cable links). In particular, it is not a reasonable assumption to neglect bit-errors over wireless channels. In addition to bit-errors, source coder robustness relates to its sensitivity to packet-loss, which may occur in the network due to overload (or, sometimes, over wireless paths due to detected bit-errors in packets that are then declared lost).

A famous result from information theory, known as the source–channel separation theorem, states that there is no loss in treating source and channel coding as two separate problems [2]. It is however important to notice that in the general case, the separation can be made without loss only in the limit of infinite delay and coding complexity; a fact that has been frequently pointed out to motivate the use of *combined* source–channel codes [3]–[7]. Many current source coders employ only error concealment to make the coder robust, while we emphasize the use of combined source–channel coding, with a focus on techniques based on channel optimized vector quantization (COVQ).

The concept of COVQ originates in [3]–[7], and the use of COVQ has since been proposed in many different contexts. However, although much has already been said on the subject,

most of the previous works have been focused towards channels with bit-errors, such as the binary symmetric channel. In contrast, the main contributions of the present work are an extension to channels with *both* bit-errors *and* bit-erasures, and a demonstration of how the new COVQ technique can be implemented to enhance the performance of a subband image coder. Related work on trellis-based source coding for the binary erasure channel can be found in [8].

II. THE BINARY SYMMETRIC ERASURE CHANNEL

We consider channels where both erasures and bit-errors occur. Such channels can arise, e.g., when information is communicated over several consecutive channels with different properties. Consider for instance the case where a packet-switched network is used for long distance communication, but local access is made through a wireless link. The network may fail to deliver a packet on time, causing packet erasures. The wireless link, on the other hand, is prone to introduce bit-errors.

A wireless link, as defined by its physical properties, modulation, non-perfect channel coding, etc., can be replaced by an equivalent discrete channel. Often this discrete channel can be modeled as a *binary symmetric channel* (BSC) with a transition probability α , corresponding to the bit-error rate (BER) of the channel. In packet loss channels, erasures come in groups of many bits. We assume however, that bit-level interleaving, using a pseudo-random spreading sequence, is implemented in such a way that all bits in a *symbol* are transmitted in different packets. This way, from a symbol perspective, independent packet losses are turned into independent bit-erasures. The resulting channel is the *binary erasure channel*, defined by the erasure probability β which will be referred to as the bit-loss rate (BLR). Note that in our case the BLR is equal to the packet loss rate (PLR).

If the binary symmetric channel and the binary erasure channel are concatenated, the *binary symmetric erasure channel* (BSEC), depicted in Fig. 1, is obtained. For the BSEC, bits are complemented with probability α and lost with probability β . Thus, the probability of a correctly received bit is $(1 - \alpha)(1 - \beta)$, the probability of a complemented bit is $\alpha(1 - \beta)$ and the probability of an erasure is β . Note that both the binary symmetric channel and the erasure channel are obtained as special cases of the BSEC.

In practical scenarios the BSEC is naturally a quite crude model. For example the channel parameters may vary during the transmission of a set of interleaved packets, since packets may follow different routes (in this case, α and β model

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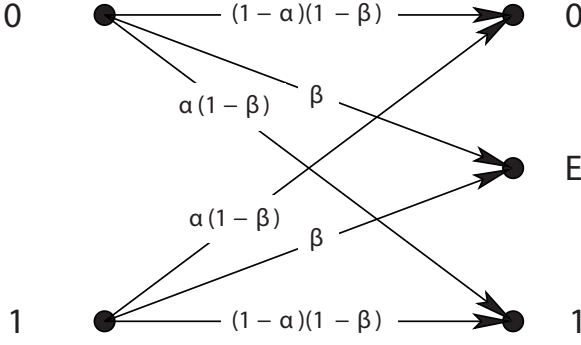


Fig. 1. Transition probabilities of the BSEC.

the average behavior). Still, the BSEC serves as a useful approximation in a scenario where both errors and losses are present. The parameters α and β can be estimated using similar techniques as those used for related Markov models for Internet transmission [9].

III. CHANNEL OPTIMIZED VQ FOR THE BSEC

While general results for COVQ are well known [4], [6], [7], some basic results are repeated here for convenience. In general, a vector quantizer (VQ) is defined by two basic operations, the encoder and decoder. The *encoder*, $\varepsilon(\cdot)$, transforms a source vector, $\mathbf{X} \in \mathbb{R}^k$, into a quantization index, $I = \varepsilon(\mathbf{X})$, $I \in \{0, 1, \dots, M-1\}$. Assuming $M = 2^L$, this corresponds to sending L bits per source vector, that is, the rate is $R = L/k$. The encoder operation is defined by a partitioning, $\mathcal{P} = \{S_0, S_1, \dots, S_{M-1}\}$, of $\mathbb{R}^k$ such that $\varepsilon(\mathbf{x}) = i$, iff $\mathbf{x} \in S_i$. The *decoder*, $\delta(\cdot)$, is a mapping from a finite set of integers to an associated set of vectors, $\mathbf{Y} = \delta(J)$, $J \in \{0, 1, \dots, N-1\}$. The set of reconstruction vectors, $\mathcal{C} = \{\mathbf{y}_0, \mathbf{y}_1, \dots, \mathbf{y}_{N-1}\}$, $\mathbf{y}_j \in \mathbb{R}^k$, is called the *decoder codebook*.

Suppose that the index $I = \varepsilon(\mathbf{X})$ is sent over a noisy channel, and that J is observed at the receiver. Assume also that the distortion measure $d(\mathbf{x}, \mathbf{y})$ associated with mapping an input vector, $\mathbf{x}$, into an output vector, $\mathbf{y}$, is given by the squared Euclidean distance, i.e.

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|^2.$$

Then necessary conditions for minimizing the expected distortion, $D(\mathcal{P}, \mathcal{C}) = E[d(\mathbf{X}, \mathbf{Y})]$, are:

$$\mathcal{S}_i = \left\{ \mathbf{x} : \sum_{j=0}^{N-1} P(j|i) \|\mathbf{x} - \mathbf{y}_j\|^2 \leq \sum_{j=0}^{N-1} P(j|i') \|\mathbf{x} - \mathbf{y}_j\|^2, \forall i' \neq i \right\} \quad (1)$$

and

$$\mathbf{y}_j = E[\mathbf{X}|J = j] = \frac{\sum_{i=0}^{M-1} \Pr(I = i) P(j|i) \mathbf{c}_i}{\sum_{i=0}^{M-1} \Pr(I = i) P(j|i)}, \quad (2)$$

where in (2) we defined the *encoder centroids* $\mathbf{c}_i = E[\mathbf{X}|I = i]$, and where $P(j|i) = \Pr(J = j|I = i)$ are the transition

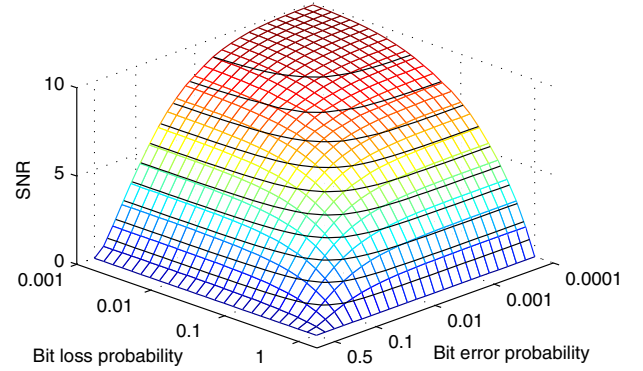


Fig. 2. Performance of COVQ over different channel parameters.

probabilities of the channel. Generally [4], [6], [7], COVQ design is based on iterating between (1) and (2) until convergence to a (local) optimum in terms of a stationary point of $D(\mathcal{P}, \mathcal{C})$.

The precise model assumed for the discrete channel influences the design only through the transition probabilities $P(j|i)$. When transmitting L bits, $b_L, \dots, b_1$, over the BSEC, with b_1 the LSB and b_L the MSB of the index i , we can define the output index j as $j = \sum_{l=1}^L d_l 3^{l-1}$, where: no error $\Rightarrow d_l = b_l$, bit-error $\Rightarrow d_l = 1 - b_l$, and erasure $\Rightarrow d_l = 2$. In this case, the transition probabilities can be computed easily, for example with $L = 2$, we get $P(7|0) = \{b_1 = 0 \rightarrow d_1 = 1 \text{ and } b_2 = 0 \rightarrow d_2 = 2\} = \alpha(1 - \beta)\beta$, and $P(8|0) = \{b_1 = 0 \rightarrow d_1 = 2 \text{ and } b_2 = 0 \rightarrow d_2 = 2\} = \beta^2$, and so on.

Note that when transmitting over the BSEC, the decoder codebook is larger than the number of encoder indexes. More specifically, $N = 3^L > M = 2^L$. This, effectively, results in *soft decision* source decoding at the receiver, c.f. [13], [14]. The soft information used by the decoder stems from the additional erasure output symbol of the BSEC. We stress that under the assumptions made and since the decoder codebook is optimal subject to these assumptions, the decoder utilizes the available soft information in an optimal manner. An alternative interpretation of COVQ's trained for the BSEC, is in terms of multiple description coding. Each received bit can be used to decrease the distortion of the reproduced value, independently of which other bits are received. Especially in the case $\alpha = 0$, $\beta > 0$ (no bit-errors, only erasures) we see that, as in multiple description quantization, each bit is carried on a "channel" that either works without error or is turned off, corresponding to descriptions (bits) which are either received or lost. Hence, training an L -bit COVQ for the BSEC is similar to designing a multiple description VQ for L channels. Related recent work uses channel optimized scalar quantizers to design multiple description coders for the two channel case [15], see also [16] for an extension to VQ and multiple channels.

The basic principle of optimizing a COVQ jointly for bit-errors and erasures, that is for $P(j|i)$'s corresponding to the channel in Fig. 1, is illustrated in Fig. 2. The figure shows the performance of COVQ's trained for uncorrelated Gaussian data with unit variance. An ordinary VQ, trained with the splitting algorithm [17] followed by the binary switching

TABLE I
VECTOR SIZES AND BIT ALLOCATION

Subband number	1	2	3	4	5	6	7	8	9	10	11	12	13
Vector size	1	2	2	2	4	4	4	4	4	4	4	4	4
VQ rate	8	10	10	9	10	10	8	8	8	4	0	0	0
COVQ rate	15	15	15	9	10	10	9	7	7	2	0	0	0

algorithm [18], is used to initialize the COVQ training. In this example, the rate is 2 bits/dimension for 3-dimensional data, and the channel is perfectly matched to the training parameters. The x - and y -axes correspond to different bit error and bit loss probabilities α and β respectively, and the z -axis shows performance in terms of reproduced signal-to-noise ratio $E\|\mathbf{X}\|^2/E[d(\mathbf{X}, \mathbf{Y})]$.

IV. A VQ-BASED SUBBAND IMAGE CODER

We have constructed a simple subband image coder using vector quantization of the image subband samples. VQ-based subband image coding can be done in a variety of ways; a good overview can be found in [19]. Our coder uses a four-level pyramid subband decomposition [20]–[22], giving a total of 13 different subbands. Vectors are formed within subbands (no crossband vectors) and each subband is assigned a certain bit rate according to the “equal-slope” method (c.f. [19]). The output indices from the VQ’s are put directly in the bit stream, without any additional entropy- or channel coding (except for bit-level interleaving), resulting in a fixed bit rate. Although better compression is possible by using variable-rate entropy coding, we motivate our fixed bit-rate structure by its avoidance of error propagation.

Designing COVQ’s for the described image coder requires the use of training data. The approach we use to generate training data is to fit a Gaussian mixture model to the empirical probability density function of vectors from each subband [23]. The model parameters are estimated from a set of images not including the test images. The estimated models can then easily be applied in generating an arbitrary amount of training data. The main reason for using a model of the source distribution instead of training on image data directly, is to avoid over-fitting the COVQ’s to a small set of image data.

V. IMAGE RESULTS

Fig. 4 shows the performance of a subband coder based on ordinary VQ compared with two COVQ-based subband coders. In all three cases the simulated channel is the BSEC with 10% erasures and 5% bit errors. The COVQ’s used in Fig. 4(b) are trained for a BSC with 5% bit errors, without knowledge about the erasures. The COVQ’s of Fig. 4(c) are trained according to our proposed scheme with full knowledge of the BSEC. For the ordinary VQ and the BSC-optimized COVQ cases, codewords received with lost bits are reproduced by the unconditional mean of all codewords that match the partially received codeword. The VQ dimension used for each subband, together with the rate allocation giving an overall rate of 0.5 bits/pixel, is given in Table I. The subband numbering starts from the lowest frequency subband.

As can be seen in Fig. 4, the COVQ-based schemes clearly outperform using ordinary VQ. The gain of training for full

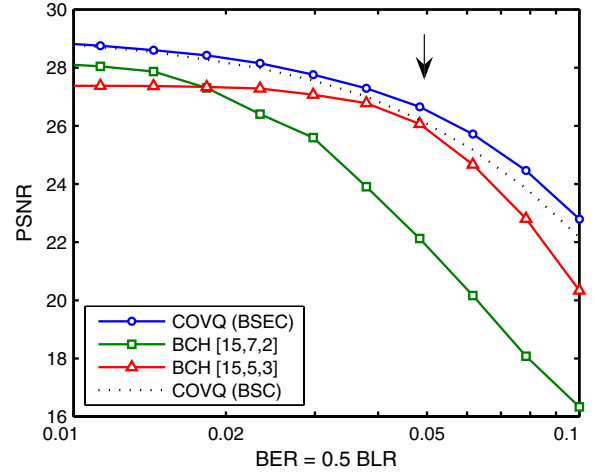


Fig. 3. Performance of COVQ compared with VQ+FEC.

knowledge of the BSEC compared with assuming bit-errors only is clearly visible, but not as striking. Still this gain comes at almost no cost in increased complexity (the only essential difference is in storing larger decoder codebooks).

VI. COMPARISON WITH FORWARD ERROR CORRECTION

To compare the performance of the COVQ approach with the performance of a more traditional one, the same image coder structure is used, but with VQ’s optimized for the source statistics only (as given by the Gaussian mixture models). The resulting bit stream from the image coder is then further encoded using BCH codes for protection against errors and erasures. Using *forward error correction* (FEC) increases the total bit rate, so to get a fair comparison, the number of bits allocated to the image coder has to be reduced until the total transmission rate is the same for COVQ and VQ+FEC.

Fig. 3 shows the performance for four different approaches. Two COVQ coders and two FEC coders. The first COVQ coder is based on our proposed scheme and is optimized for BER = 5% and BLR = 10%, this operating point is marked by an arrow in Fig. 3. The other COVQ coder is optimized for BER = 5% using a traditional COVQ design for the BSC. The performance of the two COVQ coders is compared with two FEC coders, using BCH codes of length 15 and dimensions 7 and 5 respectively. The former BCH code can correct 2 errors or 4 erasures and the latter can correct 3 errors or 6 erasures by using ML decoding. In all cases the well known test image Goldhill was encoded to a total bit rate of 0.5 bpp. To demonstrate a channel with both bit errors and erasures, the erasure probability was set to twice the bit error probability for a range of values. The results show that COVQ outperforms FEC for all the simulated channels, even though it is only



(a)



(b)



(c)

Fig. 4. Comparison of (a) subband coder using regular VQ, (b) subband coder using COVQ trained for the BSC with 5% bit-errors, and (c) subband coder using COVQ trained for the BSEC with 5% bit-errors and 10% erasures. In all three cases, the image was encoded at 0.5 bits/pixel, and a BSEC with 10% bit-losses and 5% bit-errors was simulated.

optimized for *one* specific point on the curve. The flat part of the curves for FEC corresponds to the case when the BCH code can correct *all* errors and erasures. When the code breaks down, the performance drops rapidly. COVQ shows a smoother degradation, observed by others as being typical of COVQ.

Note that it should always be possible to find a COVQ that performs equally well or better than VQ+FEC for any given channel (and a certain channel block length). Such a COVQ can be found trivially, by using VQ+FEC as the initial encoder partitioning in the COVQ training. Since training only can improve performance, one iteration is sufficient to guarantee

equally good or better performance (at the design point). Furthermore, the scheme based on COVQ will in general be more robust toward mismatch in assumptions about the channel parameters.

VII. SUMMARY

We have presented a simple but important extension of previous work on COVQ for binary memoryless channels with bit-errors to channels with both errors and erasures. Such channels are motivated by a scenario where parts of a network connection are over wireless links. Bit-errors stem from transmission errors and bit-erasures stem from packet losses. An

implicit assumption is that no retransmissions are utilized in the network, e.g., due to strong real-time requirements. Hence, packets potentially recognized to contain bit-errors are not declared lost. Instead, packet losses are assumed to be due to other phenomena, such as network overload.

The results show that COVQ is a useful tool for designing joint source-channel codes for the binary symmetric erasure channel (BSEC). In particular, advantages of joint source-channel coding over separate source and channel coding were demonstrated.

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